RESEARCH SOFTWARE / INDEPENDENT VALIDATION

TorchSim: initial findings

Five findings supported by independent reference calculations, plus three additional observations for review.

2x In one direct-force option, symmetric pair forces are doubled. Four independent libraries agree with each other and with two working TorchSim controls.

Five findings with native reference controls

Finding and possible consequence	Independent evidence
F001 / Pair forces are doubled Can change acceleration and simulated motion.	ASE, LAMMPS, JAX MD and TorchMD agree; TorchSim controls agree too.
F002 / Atomic energies do not add up Their sum is twice the total, affecting local energy analysis.	ASE preserves the allocation sum; TorchSim fails that same invariant.
F003 / Inactive interactions give invalid forces Can invalidate runs in the tested mixed-species setup.	JAX MD returns finite, zero forces for the matched configuration.
F004 / Timesteps are applied to directions Can distort motion when systems use different timesteps.	Independent ASE runs give the expected displacement per system.
F006 / Export loses custom atomic masses Can change later calculations that depend on mass.	Native ASE preserves isotope masses through its own round trip.

Three additional observations

F007: Grouped inputs ignore a requested precision change. **F008:** A rejected Monte Carlo move leaves properties from the discarded proposal. Both are reproduced consistency issues. **F005:** Unequal-mass initialization uses a different drift-removal rule from ASE; JAX MD shares TorchSim's rule, so its intended behavior needs clarification.

How to read these results

Native reference comparisons use controlled inputs, matched conventions and CPU double precision. Only F001 was compared across all five libraries. The other findings use applicable native controls or state-consistency checks. Consequences above describe plausible effects in affected paths; effects on users or published research have not been measured.